Chapter 2

Exceptions

MedTech
MEDITERRANEAN INSTITUTE OF TECHNOLOGY

Introduction

When we do programming, it is important to take into account all eventual errors.

To make a program robust, all the possible errors have to be detected and handled.

Any program may face an exceptional condition (or exception) during its execution.

There are some programming languages (ex Java) proposing a **mechanism of error handling** based on the **exception notion**.

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

What is an exception?

An exception is a **situation that stops the normal execution** of a program (not a bug)

Examples

an inexistant file which is necssary for the execution of the problem.

Division by zero

Exceeding the bounds of a table

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

What is an exception?

```java
public class ExceptionsDemo {
  public static void show() {
    sayHello( name: null);
  }

  public static void sayHello(String name) {
    System.out.println(name.toUpperCase());
  }
}
```

```
Run:  Main
/Library/Java/JavaVirtualMachines/jdk-12.0.1.jdk/Contents/Hom
Exception in thread "main" java.lang.NullPointerException
    at com.codewithmosh.exceptions.ExceptionsDemo.sayHello(Ex
    at com.codewithmosh.exceptions.ExceptionsDemo.show(Except
    at com.codewithmosh.Main.main(Main.java:7)
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB Business | MedTech Ingéniorat | LCI Langues

Exception: Definition

An exception is an object that may be issued by a method if an execptional event occurs (including the errors).

Thus, the method engendering an exception **does not return a result** but **emits an exception** and explains the cause.

The propagation of an exception emission follows the next steps:

1. An exception is generated inside a method
2. If the method has a treatment for this exception, we go to step 4, otherwse we go to step 3
3. The exception test is returned to the method calling the current method (which emits the exception), we go to step 2
4. The exception is handled and the execution of the program will continuate normally.

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Exceptions handling

The exception is spread from where it is thrown, it goes back from caller to caller and stops at every encountered exception handler.

Each exception handler is reviewed, starting from the first that is declared.

If this exception handler is compatible with the exception type, the associated code then the recovery associated code will be executed and then the program takes back normally its execution just after the last exception handler.

If there is no suitable exception handler, the same process is reiterated by going up, until an exception handler captures it or the main function is reached.

Advantages of exceptions handling

Easy programming and readability

It is possible to group the error management at the same level in order to avoid redundancies of code handling errors.

A clean and explicit error management

some programming languages use the return value of methods to report an error to the calling method. Since it is not the role of the return value to describe an error. Disassociating the return value and the exception allows the exception to accurately describe the line of code that caused the error and the nature of the error.

What is an exception?

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Unchecked exception

```java
public class ExceptionsDemo {
  public static void show() {
    sayHello( name: null);
  }

  public static void sayHello(String name) {
    System.out.println(name.toUpperCase());
  }
}
```

```
Run:  Main
/Library/Java/JavaVirtualMachines/jdk-12.0.1.jdk/Contents/Hom
Exception in thread "main" java.lang.NullPointerException
    at com.codewithmosh.exceptions.ExceptionsDemo.sayHello(Ex
    at com.codewithmosh.exceptions.ExceptionsDemo.show(Except
    at com.codewithmosh.Main.main(Main.java:7)
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB Business | MedTech Ingéniorat | LCI Langues

Checked Exceptions

```
import java.io.FileReader;

public class ExceptionsDemo {
  public static void show() {
    var reader = new FileReader( fileName: "file.txt");
  }
}
```

Unhandled exception: java.io.FileNotFoundException

Catch Blocks

```java
import java.io.FileNotFoundException;
import java.io.FileReader;

public class ExceptionsDemo {
  public static void show() {
    try {
      var reader = new FileReader( fileName: "file.txt");
      System.out.println("File opened");
    } catch (FileNotFoundException ex) {
      System.out.println("File does not exist.");
    }

  }
}
```

Parsing java... [Code]

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Catching Multiple types of Exceptions

```java
public class ExceptionsDemo {
  public static void show() {
    try {
      var reader = new FileReader( fileName: "file.txt");
      var value = reader.read();
    }
    catch (FileNotFoundException e) {
      e.printStackTrace();
    }
    catch (IOException e) {
      System.out.println("Could not read data.");
    }
  }
}
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Reversing Order of Exceptions

```
public class ExceptionsDemo {
  public static void show() {
    try {
      var reader = new FileReader( fileName: "file.txt");
      var value = reader.read();
    }
    catch (IOException e) {
      System.out.println("Could not read data.");
    }
    catch (FileNotFoundException e) {
      e.printStackTrace();
    }


  }
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Reason

java.io

Class FileNotFoundException

```
java.lang.Object
    java.lang.Throwable
        java.lang.Exception
            java.io.IOException
                java.io.FileNotFoundException
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Solution

```
public class ExceptionsDemo {
  public static void show() {
    try {
      var reader = new FileReader( fileName: "file.txt");
      var value = reader.read();
    }
    catch (IOException e) {
      System.out.println("Could not read data.");
    }
//    catch (FileNotFoundException e) {
//      e.printStackTrace();
//    }



  }
```

Combining two Exceptions

```
public class ExceptionsDemo {
  public static void show() {
    try {
      var reader = new FileReader( fileName: "file.txt");
      var value = reader.read();
      new SimpleDateFormat().parse( source: "");
    }
    catch (IOException | ParseException e) {
      System.out.println("Could not read data.");
    }
  }
}
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Finally Block

```java
public class ExceptionsDemo {
  public static void show() {
    FileReader reader = null;

    try {
      reader = new FileReader( fileName: "file.txt");
      var value = reader.read();
    }
    catch (IOException e) {
      System.out.println("Could not read data.");
    }
    finally {

      reader.close();
    }
  }
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

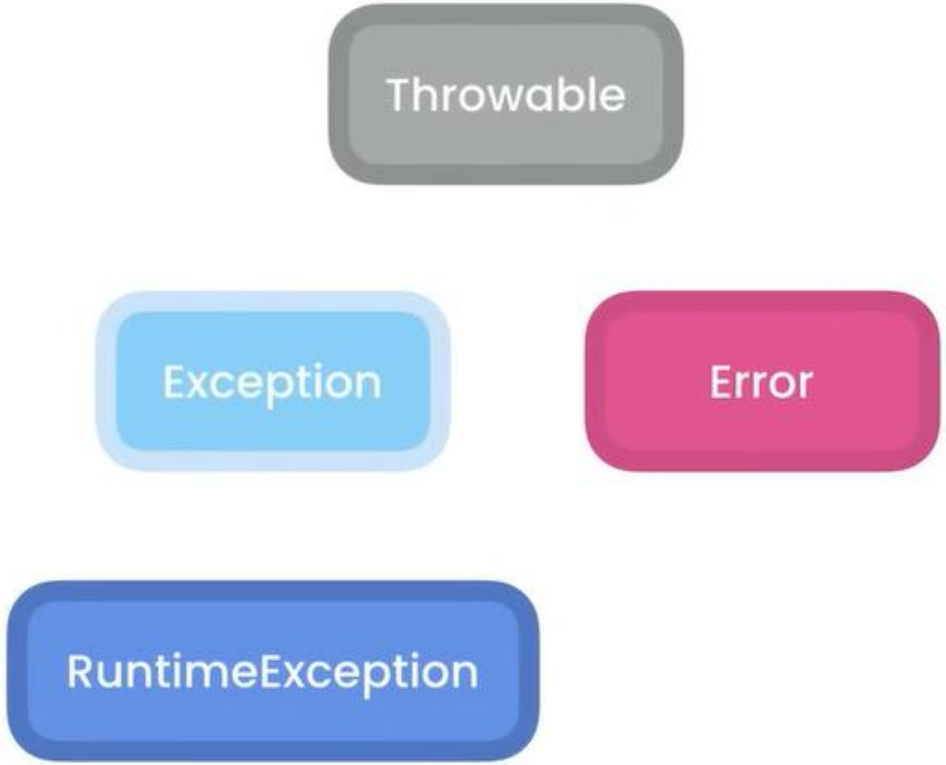
Finally Block…

```
FileReader reader = null;

try {
  reader = new FileReader( fileName: "file.txt");
  var value = reader.read();
}
catch (IOException e) {
  System.out.println("Could not read data.");
}
finally {
  if (reader != null) {
    try {
      reader.close();
    } catch (IOException e) {
      e.printStackTrace();
    }
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Exception Class Hierarchy



SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Exception declaration

It is necessary to declare for each method, the exception classes that can be eventually emitted.

This declaration is done at the end of the signature thanks to the keyword **throws.**

Exemple

```
public static int parseInt(String s) throws NumberFormatException { ...
}
```

This method converts a string (containg only numbers) to an int.

An error occurs when this string contains other types of characters. In this case an exception *NumberFormatException* will be emitted.

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Declaration – raise an exception

An exception may be emitted in/by a method

- By another method called inside the first one
- By creating an object which instantiates the class Exception (class **Throwable**) or explicitely by the keyword **throw.**

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Example

```
public class ExampleException {
/* This method returns the month according to the number provided as a parameter. If this number is
invalid (does not belong to [1, 12], an exception of type IndexOutOfBoundsException will be emitted.
*/

public static String month(int month) throws IndexOutOfBoundsException {
    if ((month < 1) || (month > 12)) {
                throw new IndexOutOfBoundsException( "the month number " + month + " should be
            between 1 and 12"); }
    if (month == 1)
        return "January";
    else if (month == 2)
        return "February";
    ...
    else if (month == 11)
        return "November";
    else
        return "December";}
}
```

The exception *IndexOutOfBoundsException* means that the provided index exceeds the maximum and minimum prefixed (between 1 and 12 for this example).

We remark that only the error detection was formulated in the previous example without its handling.

Interception et treatement of an exception

```
public class TreatementExceptionExample {
    public static void main(String[] args) {
        System.out.print("please input the month number : ");
        try {
            BufferedReader input = new BufferedReader( new InputStreamReader(System.in));
            String choice = input.readLine();
            int number = Integer.parseInt(choice);
            System.out.println(ExampleException.month(number));
        } catch (IndexOutOfBoundsException e) {
            System.err.println("incorrect number: " + e.getMessage());}
          catch (NumberFormatException e) {
            System.err.println("incorrect input: " + e.getMessage()); }
          catch (IOException e) {
            System.err.println("access error: " + e.getMessage()); }
        // we catch multiple types of exceptions here
    }
}
```

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB Business | MedTech Ingéniorat | LCI Langues

Three classes of exceptions are handled in this example:

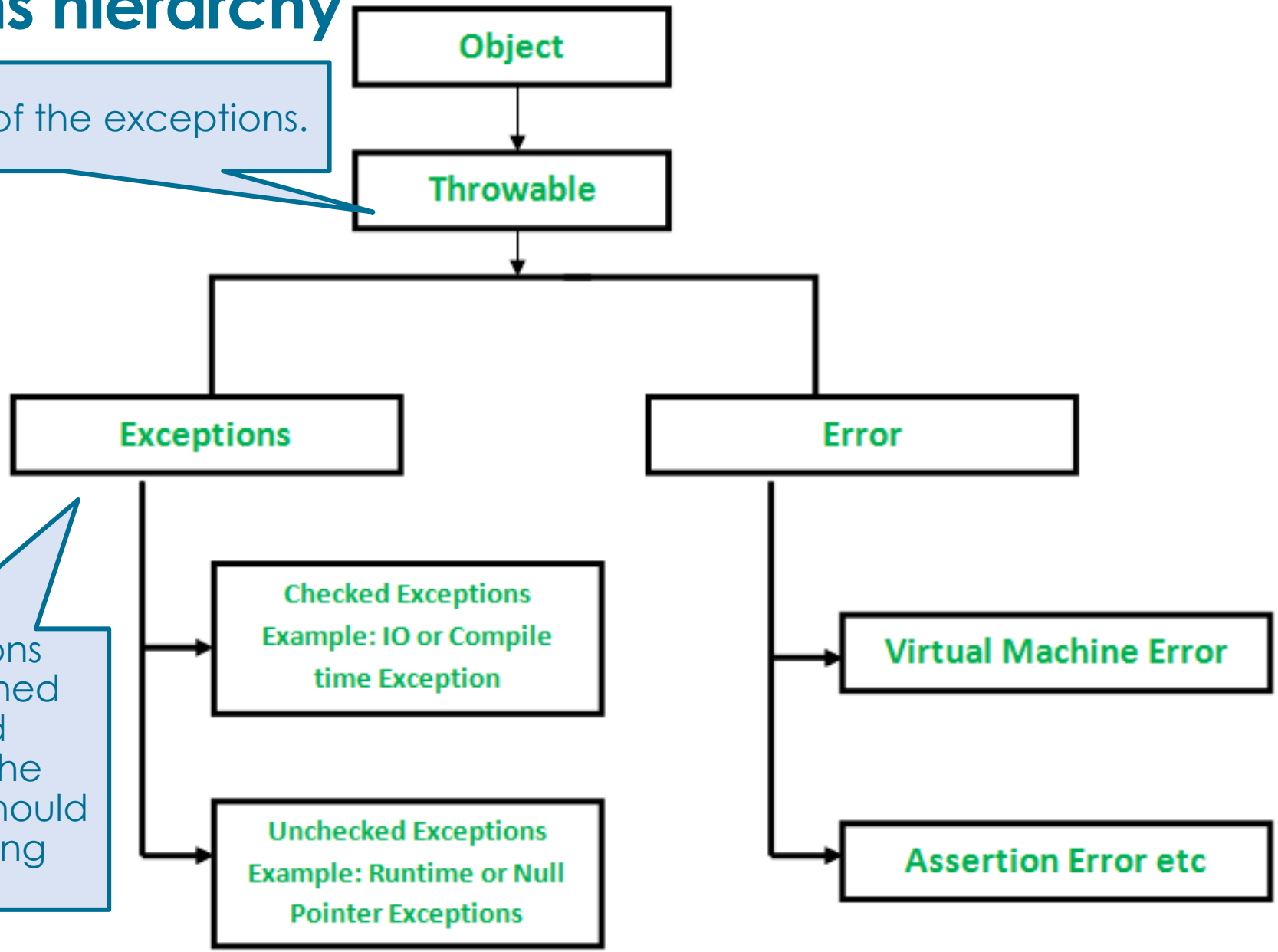
- *IndexOutOfBoundsException* (emitted by the method *month*) occurs when the number is <1 or >12
- *NumberFormatException* (emitted by the method *parseInt*) occurs when the text given by the user cannot be converted to int
- *IOException* (emitted by the method *readLine*) occurs when there is an access error in input devices.

For each case, the treatment is to display a message about the encountered exception.

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Exceptions hierarchy

The super class of the exceptions.



These exceptions should be catched and handled (**try/catch**) or the calling method should handle it by using (**throws**)

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB | MedTech | LCI
Business Ingéniorat Langues

Frequent Exception classes (some examples)

Class	Description
AWTException	Graphical Operations
`ClassCastException`	Error of incompatibility in converting an object
FileNotFoundException	attempt to open a non-existent file
`IndexOutOfBoundsException`	Error in the access to an element in a set
`IOException`	Input/output operations
`NullPointerException`	A null pointer null is received by a method not accepting this value, or when calling a method or a variable from a pointer (length or access to a null box of a table, ...)
ArrayIndexOutOfBoundsException	Access to a non-existent box in the table
StringIndexOutOfBoundsException	Access to a non-existant character of a string
`NegativeArraySizeException`	negative size of a table
`NumberFormatException`	Error in converting a string to a number

SMU
SOUTH MEDITERRANEAN UNIVERSITY
MSB Business | MedTech Ingéniorat | LCI Langues